EXPERIMENT :-5(B)

CODE:-

```
#include <stdio.h>

#define INF 9999

struct Edge {
   int u, v, w;
};

int main() {
   int n, e, source;
   int dist[20];
   int i, j;

   printf("Enter number of vertices and edges: ");
   scanf("%d%d", &n, &e);

   struct Edge edges[e];

   printf("Enter edges (u v weight):\n");

   for (i = 0; i < e; i++)
      scanf("%d%d%d",
          &edges[i].u,
          &edges[i].v,
          &edges[i].w);
```

```
printf("Enter source vertex: ");
scanf("%d", &source);

for (i = 0; i < n; i++)
   dist[i] = INF;

dist[source] = 0;

// Relax all edges n-1 times
for (i = 1; i <= n - 1; i++) {
   for (j = 0; j < e; j++) {
      int u = edges[j].u;
      int v = edges[j].v;
      int w = edges[j].w;

      if (dist[u] != INF &&
         dist[u] + w < dist[v]) {
         dist[v] = dist[u] + w;
      }
   }
}

// Check negative cycle
for (j = 0; j < e; j++) {
   int u = edges[j].u;
   int v = edges[j].v;
   int w = edges[j].w;
```

```
      if (dist[u] != INF &&
         dist[u] + w < dist[v]) {
         printf("Negative weight cycle exists!\n");
         return 0;
      }
   }

   printf("\nShortest distances from %d:\n", source);

   for (i = 0; i < n; i++)
      printf("To %d = %d\n", i, dist[i]);

   return 0;
}
```